THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	H^+ is needed for acidic behaviour.
2	**(b)** 7	Complete neutralisation to NaCl.
3	**(b)** plaster of Paris	It sets into gypsum.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Heat could cause splashing of acid.
5	**(c)** 25 mL	Ratio 1 : 2.
6	**(c)** $CuSO_4 \cdot 5\,H_2O$	Blue vitriol.
7	**(a)** Both A and R are true and R is the correct explanation of A.	No free ions are produced.
8	**(a)** 5	H^+ falls tenfold so pH rises by 1.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Without water the acid does not ionise, so no H^+ is available to change the litmus; R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the disinfecting action is due to the chlorine it releases, not due to the way it is manufactured.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	CO_2 (1); $CaCO_3 + 2\,HCl \longrightarrow CaCl_2 + H_2O + CO_2$, lime water turns milky (1).	1+1
12	Plaster of Paris 145 g (1); Water lost 27 g (1).	1+1
13	Dilution is exothermic (1); Water floats and boils, spraying acid or cracking the glass (1).	1+1
14	pH becomes 5 (1); Still acidic (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	pH 10 and 13 turn it pink (1); pH 1 and 4 neutralise a base (1); Acidity order: pH 1 > 4 > 7 > 10 > 13 (1).	1+1+1
16	X is $CaCO_3$, Y is CaO (1); $CaCO_3 \longrightarrow CaO + CO_2$ (1); $CaO + H_2O \longrightarrow Ca(OH)_2$ (1).	1+1+1
17	Acids react with copper/brass (1); Toxic salts form (1); Glass does not react with acids (1).	1+1+1
	OR: Add quick lime or slaked lime (1); Basic oxide neutralises soil acidity (1); Soil pH approaches neutral, improving crops (1).	1+1+1
18	Strong acid has lower pH (1); Weak acid ionises partially (1); Universal indicator shows red vs orange/yellow (1).	1+1+1
	OR: Methanoic acid is injected (1); Rub baking soda (1); Or use leaf of dock plant (mild base) (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Sample Q (pH = 7) (1). (ii) Sample S (highest pH) (1). (iii) R and S (pH greater than 7) (1). (iv) Quick lime / slaked lime / chalk (any one) (1), because it is basic and neutralises the excess acid in the soil (1).	1+1+1+2
20	First: neutral (pH 7) (1); Equal amounts of acid and base (1); Second: acid in excess (1); pH less than 7 (1); Excess H^+ makes the solution acidic (1).	1+1+1+1+1
	OR: Enzymes need a narrow pH (1); Stomach acid (pH about 1.4) helps digestion (1); Mouth must stay above 5.5 (1); Excess acid causes indigestion; antacids help (1); Acid in mouth produces tooth decay (1).	1+1+1+1+1

Mistakes to avoid:

- Pouring water into concentrated acid; always add the acid slowly to water with stirring because the process is highly exothermic.
- Writing that a solution with pH 3 is 'less acidic' than one with pH 5; a lower pH means a **stronger** acid.
- Mixing up the formulae: baking soda is $NaHCO_3$, washing soda is $Na_2CO_3 \cdot 10\,H_2O$, plaster of Paris is $CaSO_4 \cdot \frac{1}{2}\,H_2O$ and bleaching powder is $CaOCl_2$.

- Thinking that all compounds containing hydrogen are acids; glucose and alcohol do not release H^+ ions.
- Writing H_2 as the gas formed when an acid reacts with a carbonate; the gas is CO_2.
- Assuming neutralisation always gives a neutral solution; a strong acid with a weak base gives an acidic salt solution (and vice-versa).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** The first is more acidic	Lower pH means a more acidic solution.
2	**(c)** $NaHCO_3$	It decomposes to sodium carbonate.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Absorbs water to form gypsum.
4	**(c)** 145 g	172 - 27 = 145 g.
5	**(c)** chlorine	Moist chlorine bleaches litmus after reddening.
6	**(c)** A is true but R is false.	It is white; blue is the hydrated salt, so R is false.
7	**(b)** 36%	90/249.5 = 36%.
8	**(d)** Q and R	pH 5 and pH 9 are equally far from 7 on opposite sides, so their effects cancel.
9	**(c)** A is true but R is false.	Enamel (calcium phosphate) is corroded below pH 5.5 by acids, formed when bacteria degrade sugar; R is false.
10	**(d)** A is false but R is true.	Washing soda is $Na_2CO_3 \cdot 10\,H_2O$, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Oxides of sulphur and nitrogen dissolve in rain (1); Form strong acids such as H_2SO_4 (1).	1+1
12	5 mL : 20 mL ratio (1); 25 mL needs 100 mL (1).	1+1
13	It is a salt of a strong acid and a weak base (1); NH_4^+ produces H^+ on hydrolysis (1).	1+1
14	It is a salt of strong base and weak acid (1); pH about 8.3, mildly basic (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$ (1); CO_2 makes cake spongy (1); Na_2CO_3 left is bitter, so tartaric acid is added to remove it (1).	1+1+1
16	Pure water has very few ions (1); Acid ionises to give H^+ and anions (1); These mobile ions carry current (1).	1+1+1
17	(i) washing soda $Na_2CO_3 \cdot 10\,H_2O$ (1); (ii) plaster of Paris $CaSO_4 \cdot \frac{1}{2}\,H_2O$ (1); Both are hydrated salts (1).	1+1+1
	OR: Molar mass 172 (1); Water 36 (1); 36/172 x 100 = 20.9% (1).	1+1+1
18	Neutralisation is exothermic (1); $NaOH + HCl \longrightarrow NaCl + H_2O + \text{heat}$ (1); Heat appears due to formation of water (1).	1+1+1
	OR: Acid rain lowers river pH (1); Fish and other life cannot survive at low pH (1); Food chains are affected (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Chlor-alkali process (1). (ii) X is chlorine, Y is hydrogen (1). (iii) Z is sodium hydroxide solution (1). (iv) $2\,NaCl + 2\,H_2O \longrightarrow 2\,NaOH + Cl_2 + H_2$ (1); $Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$ (1).	1+1+1+2
20	$NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3$ (1); $2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$ (1); $Na_2CO_3 + 10\,H_2O \longrightarrow Na_2CO_3 \cdot 10\,H_2O$ (1); Uses: baking soda for cakes/antacid (1); Washing soda for cleaning and softening hard water (1).	1+1+1+1+1
	OR: HCl: litmus red, phenolphthalein colourless, methyl orange red (1); NaOH: litmus blue (1); NaOH: phenolphthalein pink, methyl orange yellow (1); NaCl: litmus unchanged (1); NaCl: phenolphthalein colourless, methyl orange orange (1).	1+1+1+1+1

Mistakes to avoid:

- Pouring water into concentrated acid; always add the acid slowly to water with stirring because the process is highly exothermic.
- Writing that a solution with pH 3 is 'less acidic' than one with pH 5; a lower pH means a **stronger** acid.

- Mixing up the formulae: baking soda is $\mathrm{NaHCO_3}$, washing soda is $\mathrm{Na_2CO_3 \cdot 10\,H_2O}$, plaster of Paris is $\mathrm{CaSO_4 \cdot \tfrac{1}{2}\,H_2O}$ and bleaching powder is $\mathrm{CaOCl_2}$.
- Thinking that all compounds containing hydrogen are acids; glucose and alcohol do not release $\mathrm{H^+}$ ions.
- Writing $\mathrm{H_2}$ as the gas formed when an acid reacts with a carbonate; the gas is $\mathrm{CO_2}$.
- Assuming neutralisation always gives a neutral solution; a strong acid with a weak base gives an acidic salt solution (and vice-versa).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** acid in the sting is neutralised by a mild base	Neutralisation.
2	**(a)** Both A and R are true and R is the correct explanation of A.	No hydrolysis occurs.
3	**(a)** 84 g	2 x 84 g gives 44 g CO_2; 22 g needs 84 g.
4	**(d)** it is a mild base that neutralises the excess acid	$Mg(OH)_2$ is mildly basic.
5	**(a)** Both A and R are true and R is the correct explanation of A.	One unit change gives tenfold change in H^+.
6	**(c)** neutral	$H^+ = 10^{-2}$ and $OH^- = 10^{-2}$ neutralise.
7	**(c)** baking soda	$NaHCO_3 + HCl$ gives CO_2.
8	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true but R does not explain; strength depends on ionisation.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Both contain hydrogen but do not release H^+ or other ions, so they are not acids and do not conduct.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the sponginess is due to CO_2 released when the mixture is heated or moistened. R only states the composition.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	CO_3^{2-} hydrolyses more strongly than HCO_3^- (1); So pH of Na_2CO_3 about 11 and $NaHCO_3$ about 8 (1).	1+1
12	Mass of $10\,H_2O$ = 180 g (1); Out of 286 g (1).	1+1
13	Cu2+ ions hydrate in water (1); Hydrated copper ion is blue (1).	1+1
14	Baking soda decomposes on heating (1); So it is not baking soda; it is Na_2CO_3 or a carbonate (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$: 168 g gives 44 g CO_2 (1); 2.1 g is 1/80 of 168 g (1); CO_2 = 44/80 = 0.55 g (1).	1+1+1
16	(i) hydrogen from metals like Zn (1); (ii) CO_2 from carbonates (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$; $Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2$ (1).	1+1+1
17	Distilled water has no ions (1); Tap water contains dissolved salts/gases (1); These can make it slightly acidic or basic (1).	1+1+1
	OR: Neutrality requires equal moles of H^+ and OH^- (1); Concentrations may differ (1); Or strength of acid and base may differ (1).	1+1+1
18	Anode: chloride ions give Cl_2 (1); Cathode: H_2 forms (1); OH^- ions remain with Na^+: NaOH (1).	1+1+1
	OR: It absorbs CO_2 and moisture (1); Chlorine is released and activity falls (1); So it should be stored in a closed container (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Plaster of Paris (1). (ii) $CaSO_4 \cdot \frac{1}{2}\,H_2O$ (1). (iii) Gypsum, $CaSO_4 \cdot 2\,H_2O$ (1). (iv) It absorbs moisture and sets into a hard mass of gypsum (1); $CaSO_4 \cdot \frac{1}{2}\,H_2O + \frac{3}{2}\,H_2O \longrightarrow CaSO_4 \cdot 2\,H_2O$ (1).	1+1+1+2
20	NaCl electrolysed gives NaOH, Cl_2, H_2 (1); $Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$ (1); Cl_2 also used for HCl and PVC (1); NaOH used in soaps and paper (1); H_2 with Cl_2 gives HCl gas (1).	1+1+1+1+1
	OR: Above 373 K all water is lost (1); $CaSO_4$ anhydrous forms (1); It does not set with water (1); Dead burnt plaster (1); So temperature must be controlled (1).	1+1+1+1+1

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